P. Jeeva (192412636)

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI-602105

1. Rose is Botanist who is studying the Iris genus, she has collected the data of different the sepal length, sepal width, petal length, and petal width of various Iris flowers and wants to classify the flowers into their respective species based on their physical characteristics.
   Rose decides to compare the performance of different machine learning algorithms for this task. She splits her data into a training set and a test set and trains several models, including Decision tree classifier, Logistic Regression, KNN classifier. Julia wants to the performance measures based on accuracy and speed of execution. Help her do the comparison of the classification algorithms

2. For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm to output a description of the set of all hypotheses consistent with the training examples.

Example	Citations	Size	In Library	Price	Editions	Buy
1	Some	Small	No	Affordable	Few	No
2	Many	Big	No	Expensive	Many	Yes
3	Many	Medium	No	Expensive	Few	Yes
4	Many	Small	No	Affordable	Many	Yes

3. Develop a Python code for implementing Polynomial regression and show its performance
4. Develop a Python code for implementing the KNN algorithm with an example.

... [2], [3], [4], [5]].

y = [0, 0, 1, 1, 1]

model = K neighbours classifier (K = 3).

model . fit (x, y)

Predict (model. Predict ([[3]], [0]).

Input :

1 -> 0

2 -> 0

3 -> 1

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P. Jeeva , Course Code : ITA0609

4. Aim:

To write and implement the python code for KNN algorithm

Algorithm:

1. Import the data.
2. Give input and K value
3. Create the KNN model
4. Train the KNN model
5. Predict the result
6. Display the result.

Program:

```
from sklearn.neighbours ~~classifier~~ import Kneighbours classifier
  x = [[1], [2], [3], [4], [5]].
  y = [0, 0, 1, 1, 1]
model = Kneighbours classifier (k= 3).
model.fit(x,y)
Predict (model.Predict ([[3]], [0].)).
```

Input:

1 -> 0
2 -> 0
3 -> 1
4 -> 1
5 -> 1.

Given: K = 3

Output:

K=3 -> 1

③. Aim :

To write and implement the python code for poly regression.

Algorithm

1. Import the data
2. Give input data values
3. create the model
4. Train the model
5. Predict the result
6. Display the result.

Program :

```
from sklearn.polynomial_model import Polynomial regression
x = [[1], [2], [3], [4], [5]]
y = [1, 4, 9, 16, 25]
model = Polynomial regression()
model.fit(x,y)
Predict("Performance:", model.Predict([[4]], y)
```

Input :

1 -> 1

2 -> 4

3 -> 9

4 -> 16

5 -> 25

output :

6 -> 36

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Additional Sheet No. 1

(2) Aim:

To write and implement the python code for candidate elimination algorithm

Algorithm:

1. Import the .csv file
2. Give input.
3. Create a model
4. Train the model
5. Predict the result
6. Display the result.

Program:

```
data = [
        ['Some', 'Small', 'Affordable', 'Few', 'No'],
        ['Many', 'Big', 'Expensive', 'Many', 'Yes'],
       ['Many', 'Medium', 'Expensive', 'Few', 'Yes'],
      ['Many', 'Small', 'Affordable', 'Many', 'Yes']].
hypothesis h = ["?", '?', '?', '?', '?', '?']
for the rows in [4]:
        h[4] == Yes:
      Print "yes"
    if h[4] = = h[i]:
     Print rows = h[i]:
   elif h[4] != h[i]:
     Print rows = "?"
```

Input :
[['Some', 'small', 'Affordable', 'Few', 'No']
['Many', 'Big', 'Expensive', 'Many', 'Yes']
['Many', 'Medium', 'Expensive', 'Few', 'Yes']
['Many', 'small', 'Affordable', 'Many', 'Yes']]

Output :
('Many', '?', '?', 'Many', 'Yes').

① Aim :

To write and implement the python code for comparison of different algorithms.

Algorithm :

1. Import the data
2. Give input data values
3. Create the model
4. Compare the model
5. Predict the result
6. Display the result.

Program :

```
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTree_classifier.
from sklearn.linear_model import Logistic Regression.
from sklearn.neighbour_classifier import Kneighbours_classifier
data = load_iris
x = [[1], [2], [3], [8], [9], [10]]
y = [0, 0, 0, 1, 1, 1]
```

del = DecisionTreeClassifier( )
model . fit (x,y)
model = LogisticRegression ( )
model . fit (x,y)
model = Kneighbours classifier ( )
model . fit (x,y).
predict ("flowers:" model predict ([[6]], [0])

Input:

1 → 0
2 → 0
3 → 0
8 → 1
9 → 1
10 → 1

output:

6 → 1.